

Report: AI Product Intelligence System

Overview

This project uses CLIP (Contrastive Language-Image Pretraining) embeddings to solve three e-commerce challenges using a unified multimodal approach:

1. Smart Product Recommendation
2. Unique Product Catalog Creation
3. Reverse Product Search

A single embedding pipeline is reused across all tasks, making the system computationally efficient while maintaining semantic understanding of both images and text.

Methodology

CLIP was used as the core model because it maps images and text into the same embedding space. This allows similarity comparisons between products as well as between text queries and product images.

Embedding Model: CLIP ViT-B/32

Similarity Metric: Cosine Similarity

Clustering Method: Agglomerative Clustering

Task 1: Smart Product Recommendation Engine

Objective - Recommend complementary products rather than visually similar products.

Approach

A category relationship mapping was created to define complementary product pairs. For example:

- Shirts → Trousers, Belts, Shoes, Ties
- Shoes → Socks, Backpacks, Watches

After identifying complementary categories, CLIP similarity is used to rank products within those categories based on visual and semantic relevance.

Result

The system successfully generated recommendations that complement the input product instead of simply returning products from the same category.

Output

The figure below shows that for a given shirt, the system identified relevant complementary products across different categories and ranked them using CLIP-based similarity, demonstrating context-aware product recommendations.

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DEMO: Recommendations for a product in category 'Shirts'

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Top Recommendations:

- Denizen Women Blue Jeans [Jeans] – score: 0.784
- Scullers Men Scul Beige Trousers [Trousers] – score: 0.784
- Elle Women Khaki Trousers [Trousers] – score: 0.778
- Madagascar3 Girls White Shorts [Shorts] – score: 0.776
- Pieces Women Multi-Colored Belt [Belts] – score: 0.775

Task 2: Unique Product Catalog Creation

Objective - Identify duplicate and near-duplicate products to reduce catalog redundancy.

Approach

CLIP embeddings were generated for all product images. Similarity-based clustering was then applied to group products with highly similar visual representations. Products within the same cluster were treated as duplicate listings.

A similarity threshold was selected experimentally to balance duplicate detection and product diversity.

Result

The system successfully grouped visually similar products and generated a cleaner catalog containing unique product entries.

Output



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FINAL UNIQUE PRODUCT CATALOG (first 20 entries)
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	cluster_id	id	productDisplayName	articleType	baseColour	gender
0	49	50019	Hanes Women Grey Comfortsoft Waistband Cotton ...	Leggings	Grey Melange	Women
1	123	17368	Arrow Men Formal Maroon Tie+Cufflink+Pocket sq...	Accessory Gift Set	Maroon	Men
2	144	57701	Lotus Herbals Papayablem Papaya-N-Saffron Anti...	Face Moisturisers	Cream	Women
3	26	37328	Casio Sheen Women Silver-Toned Dial Chronograp...	Watches	Silver	Women
4	194	15903	Warner Bros Kids Unisex TJ Stud Sandal White C...	Casual Shoes	White	Unisex
5	92	15219	ADIDAS Originals Men Adiracer Remodel Black Ca...	Casual Shoes	Black	Men
6	122	11459	United Colors of Benetton Men Light Winters Bl...	Shirts	Blue	Men
7	177	17796	Chhota Bheem Kids Boy I Like to be Naughty Gre...	Tshirts	Green	Men
8	151	15902	Warner Bros Kids Unisex TJ Cheese Flops Purple...	Flip Flops	Purple	Unisex
9	225	51325	Wildcraft Unisex Black & Grey Checked Bishop S...	Handbags	Black	Unisex
10	19	14794	United Colors of Benetton Women Solid Maroon H...	Handbags	Maroon	Women
11	146	11403	United Colors of Benetton Women Light Winter R...	Tops	Red	Women
12	18	55273	Lakme Absolute White Intense Medium to Dark Co...	Concealer	Skin	Women
13	27	48508	French Connection Women Navy Blue & Cream Stri...	Tops	Navy Blue	Women
14	159	6055	UCB Men's Polo Neck With Tipping Collar Blue T...	Tshirts	Blue	Men
15	147	14621	United Colors of Benetton Men Stripes White Ts...	Tshirts	White	Men
16	132	57910	Fruit of the Loom Men Navy Blue Pack of Three ...	Socks	Navy Blue	Men
17	86	13947	United Colors of Benetton Women Solid Blue Tsh...	Tshirts	Blue	Women
18	200	37188	United Colors of Benetton Boys Printed Red T-s...	Tshirts	Red	Boys
19	126	24943	United Colors of Benetton Kids Girls White T-s...	Tshirts	White	Girls

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Total unique products in catalog: 255
vs original: 500 products
Catalog reduction: 245 duplicates removed
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The figure above shows cleaned product catalog generated after duplicate removal, showing representative products retained from each cluster.

Task 3: Reverse Product Search

Objective - Enable users to search products using natural language descriptions.

Approach

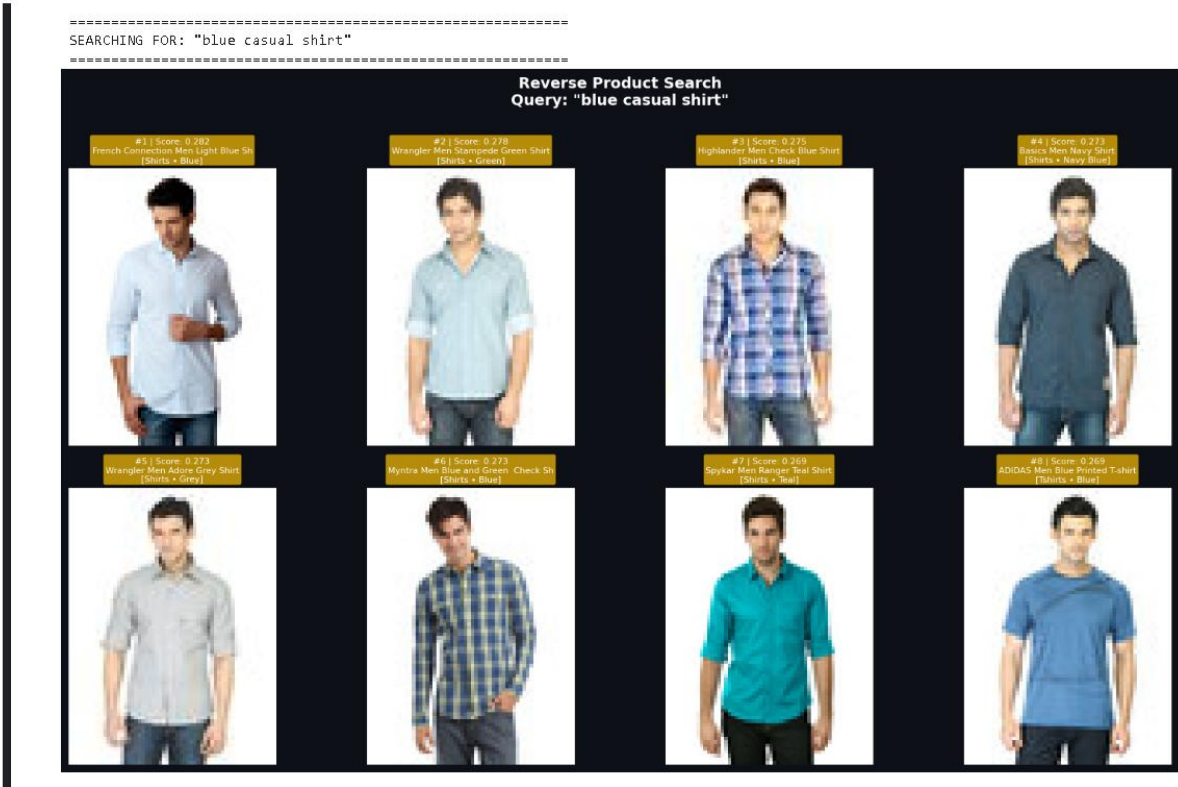
The user's text query is converted into a CLIP text embedding and compared against product image embeddings using cosine similarity. Products with the highest similarity scores are returned as search results.

Result

The system accurately retrieved relevant products for natural language queries.

Output

The figure below shows reverse product search results for the query "blue casual shirt", showing the top-ranked products retrieved using CLIP text-image similarity.



Top 8 Results for: "blue casual shirt"

Rank	Score	Category	Product Name
1	0.2822	Shirts	French Connection Men Light Blue Shirt
2	0.2778	Shirts	Wrangler Men Stampede Green Shirt
3	0.2750	Shirts	Highlander Men Check Blue Shirt
4	0.2730	Shirts	Basics Men Navy Shirt
5	0.2728	Shirts	Wrangler Men Adore Grey Shirt
6	0.2727	Shirts	Myntra Men Blue and Green Check Shirt
7	0.2694	Shirts	Spykar Men Ranger Teal Shirt
8	0.2692	Tshirts	ADIDAS Men Blue Printed T-shirt